

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 11

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 12 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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E&M Tutorial Sheet 11: Towards electromagnetism

- 1) (Revision) Calculate the self-inductance of a solenoid of length l with N turns.
- 2) Write out and then learn the four Maxwell equations in their integral form. Define all the symbols that you use.
- 3) A coaxial cable consists of two conducting concentric cylindrical shells radius a and b . The cable is long, with length l such that end effects can be ignored.
 - a) By considering charges $+Q$ on the inner cylinder and $-Q$ on the outer cylinder, calculate the potential difference between the cylinders and hence the capacitance per unit length of the cable.
 - b) Show that the energy stored in the electric field is equal to $\frac{1}{2}C\phi^2$.
 - c) By considering currents $+I$ on the inner cylinder and $-I$ on the outer cylinder, calculate the magnetic flux linked between the cylinders and hence the inductance per unit length of the cable.
 - d) Show that the energy stored in the magnetic field is equal to $\frac{1}{2}LI^2$.
 - e) Calculate an expression for $\sqrt{\frac{L}{C}}$. What units does this expression have? Evaluate the expression for $b = 2.3 \text{ mm}$, $a = 1 \text{ mm}$.

Hint on units: try to find the simplest possible SI form for the unit by considering the equations for the potentials across an inductor and a capacitor, $\phi_L = L \frac{dI}{dt}$ and $\phi_C = \frac{Q}{C}$ and remembering Ohm's Law $\phi = IR$.

PHYS10302 Vibrations and Waves

Week 11 Questions

1. X-rays of wavelength $1.54 \text{ \AA}$ are incident on a powdered sample of rock salt. The largest d -spacing for planes of ions is $3.26 \text{ \AA}$. Calculate the smallest scattering angle for Bragg scattering of the X-rays.
2. The first three X-ray diffraction peaks observed in powdered tungsten are seen at scattering angles of 40.2° , 58.3° and 73.3° for an X-ray wavelength of $1.54 \text{ \AA}$. Calculate the d -spacings that correspond to these scattering angles and find the ratios between them.

(Hint: in simple cases, the d -spacing ratios are likely to involve small integers or the square roots of integers).

3. You direct a beam of 0.154-nm X-rays at certain planes of a silicon crystal. As you increase the angle of incidence of the beam from zero, the first strong interference maximum occurs when the beam makes an angle of 34.5° with the planes.
 - (a) How far apart are the planes?
 - (b) Will you find other interference maxima from these planes at greater angles of incidence?
4. The speed of light in a vacuum is given by

$$v = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \text{constant},$$

and in a dielectric material the velocity by

$$v = \frac{1}{\sqrt{\epsilon \mu}}.$$

- (a) Find an expression for the refractive index $n = c/v$ in terms of the relative permittivity $\epsilon_r = \epsilon/\epsilon_0$ and the relative permeability $\mu_r = \mu/\mu_0$.
- (b) For most materials, μ_r is constant while ϵ_r varies with frequency. By writing an expression for the velocity in the dielectric material as a function of ϵ_r , derive an expression for the group velocity in terms of $d\epsilon_r/d\lambda$. What sign of $d\epsilon_r/d\lambda$ corresponds to anomalous dispersion? For your answer you should recall the result derived in the lectures,

$$v_g = v_p - \lambda \frac{dv_p}{d\lambda}.$$

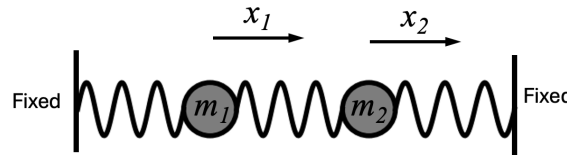
PHYS10372 Mathematics 2: Tutorial-11 Questions

1. Write the quantum mechanical linear operators, $\hat{A}$ & $\hat{B}$, as matrices where:

$$\hat{A} = |0\rangle\langle 1| + |1\rangle\langle 0|, \text{ and } \hat{B} = 5|0\rangle\langle 0| + 2|0\rangle\langle 1| + 2|1\rangle\langle 0| + 5|1\rangle\langle 1|.$$

Find the eigenstates of $\hat{A}$ & $\hat{B}$, and express them in bra-ket notation as normalised states.

2. Three springs with the same spring constant, k , are attached to masses of mass $m_1 = m$ and $m_2 = 2m$, and two fixed supports, as shown below.



The masses are displaced from equilibrium and oscillate on a frictionless, horizontal surface. Assuming the displacements from equilibrium, x_i , are given by $A_i e^{i\omega t}$, show that the equations of motion of the two masses are defined by the eigenvalues equation:

$$B\mathbf{x} = \lambda\mathbf{x} \text{ where } \mathbf{x} = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 4 & -2 \\ -1 & 2 \end{pmatrix} \text{ and } \lambda = \frac{2m\omega^2}{k}.$$

Solve the eigenvalue equation to determine the two normal-mode frequencies and find the eigenvectors. Describe the motion of the masses in the two modes.

3. By forming an eigenvalue matrix equation, find the general solution to the following coupled, first order differential equation:

$$\begin{aligned} \dot{x}_1 &= 2x_1 + 2x_2 + x_3 \\ \dot{x}_2 &= x_1 + 3x_2 + x_3 \\ \dot{x}_3 &= x_1 + 2x_2 + 2x_3 \end{aligned}$$

subject to the boundary conditions: $x_1(0) = 1, x_2(0) = x_3(0) = 0$.

4. Trade tariffs imposed on a country by a single country are defined such that a country is in one of three trading states: low-tariff, medium-tariff, high-tariff with respect to the country imposing the tariff. The transition probability between these states is defined by the matrix, A , where

$$A = \begin{pmatrix} 0.85 & 0.15 & 0.05 \\ 0.10 & 0.70 & 0.20 \\ 0.05 & 0.15 & 0.75 \end{pmatrix}$$

and the element in column- i , row- j is the probability of transitioning from state- i to state- j e.g. 0.10 represents the probability of transitioning from a low-tariff state to a medium-tariff state. We have a set of countries each initially in a low-tariff state, i.e. defined by the vector x_0 , where $x_0 = (1.0, 0.0, 0.0)^T$.

Successive decisions are made which are defined by the transition matrix A and the x_0 state evolves as a Markov chain. Determine the steady-state vector. Express your answer as fractions.

After very careful consideration, a new, trade-tariff transition probability matrix, B is implemented where

$$B = \begin{pmatrix} 0.60 & 0.10 & 0.02 \\ 0.25 & 0.65 & 0.08 \\ 0.15 & 0.25 & 0.90 \end{pmatrix}.$$

Determine what fraction of countries are now in a high-tariff state in the steady-state when evolving again from an initial state: $(1.0, 0.0, 0.0)^T$ with this new trade-tariff policy. What is the ratio of countries in the high-tariff state in the steady-state in the two policy scenarios.